NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Which of these are propositions? What are the truth values of those that are propositions?

References

Section Break Difficulty: Easy

1. Award: **3.12 points**

Do not pass go.

The given statement is not a proposition, and the truth value does not exist.

References

Worksheet Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider the given propositions.

References

Section Break Difficulty: Easy

2. Award: **3.12 points**

Mei has an MP3 player.

Identify the negations of the given proposition. (Check all that apply.)

- [ ] it is not the case that Mei has no MP3 player.
- [x] it is not the case that Mei has an MP3 player.
- [x] Mei has no MP3 player.
- [ ] Mei has no CD player.

References

Check All That Apply Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider the following propositions *p* and *q*.
p: "Swimming at the New Jersey shore is allowed."
q: "Sharks have been spotted near the shore."

References

Section Break Difficulty: Easy

3. Award: **3.12 points**

Identify the compound proposition of $\neg q \vee p$.

- ○ Swimming at the New Jersey shore is not allowed, or sharks have been spotted near the shore.

- Swimming at the New Jersey shore is allowed, or sharks have not been spotted near the shore.
- Swimming at the New Jersey shore is not allowed, and sharks have not been spotted near the shore.
- Swimming at the New Jersey shore is not allowed, and sharks have been spotted near the shore.

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Determine whether the biconditional is true or false.

References

Section Break Difficulty: Easy

4.

Award: **3.12 points**

$1 + 1 = 3$ if and only if monkeys can fly.

- True
- False

References

True / False Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Determine whether the given conditional statements is true or false.

References

Section Break Difficulty: Easy

5. Award: **3.12 points**

If $1 + 1 = 3$, then dogs can fly.

- True
- False

References

True / False Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Write the given sentence in the "if *p*, then *q*" form.

References

Section Break Difficulty: Easy

6. Award: **3.12 points**

To be a citizen of this country, it is sufficient that you were born in the United States.

- If you are a citizen of this country, then you are in the United States.
- If you were born in the United States, then you are a citizen of this country.
- If you are a citizen of this country, then you were born in the United States.
- If you were born a citizen of this country, then you are in the United States.
- If you were born in the United States, then you are in this country.

References

Multiple Choice Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Fill the provided truth tables and check whether the given conditional statements are a tautology or contradiction or contingency.

References

Section Break Difficulty: Medium

7.

Award: **3.12 points**

Is the conditional statement $\neg (p \to q) \to p$ a tautology?

- Yes
- No

References

Yes / No Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider the following argument.
If George does not have eight legs, then he is not a spider.
George is a spider.
Therefore George has eight legs.

References

Section Break Difficulty: Easy

8.

Award: **3.12 points**

Can we conclude that the conclusion is true if the premises are true?

- Yes
- No

References

Yes / No Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider the statement $n^2 + 1 \geq 2^n$ where *n* is an integer in $[1, 4]$.

References

Section Break Difficulty: Easy

9.

Award: **3.12 points**

Identify the *n* values for which the equation is to be verified in order to prove the given statement.

(You must provide an answer before moving to the next part.)

- 1, 2, 3, 4, 5
- 2, 3
- 1, 2, 3, 4
- 1, 2, 3, 5

References

Multiple Choice Difficulty: Easy

10.

Award: **3.12 points**

Identify the correct proof and the type of proof of the statement that there is a positive integer that equals the sum of the positive integers not exceeding it.

- If the sum of the positive integers less than a positive integer is not equal to the positive integer, then the sum will exceed the positive integer. This is a nonconstructive proof, because the statement is proved by contradiction.
- If the sum of the positive integers less than a positive integer is not equal to the positive integer, then the sum will exceed the positive integer. This is a constructive proof, because the statement is proved by contradiction.
- The number 1 has this property, since the only positive integer not exceeding 1 is 1 itself, and therefore the sum is 1. This is a constructive proof, because an example for which the statement is true is given.
- The number 1 has this property, since the only positive integer not exceeding 1 is 1 itself, and therefore the sum is 1. This is a nonconstructive proof, because an example for which the statement is true is given.

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Use set builder notation to give a description of each of these sets.

References

Section Break Difficulty: Easy

11.

Award: **3.12 points**

$\{-3, -2, -1, 0, 1, 2, 3\}$

- $\{x| -3 < x < 3$, where the domain is a set of integers}
- $\{x| -3 < x \leq 3\}$
- $\{x| -3 \leq x \leq 3$, where the domain is a set of real numbers}
- $\{x \in Z| -3 \leq x \leq 3\}$

References

Multiple Choice Difficulty: Easy

12.

Award: **3.12 points**

Among the following pairs of sets, identify the ones that are equal. (Check all that apply.)

- [ ] {1,3, 3, 3, 5, 5, 5, 5, 5}, {5, 3, 1}
- [ ] 1,{1,{1}}
- [ ] ∅, {∅}
- [ ] {1, 2}, { {1}, {2} }

References

Check All That Apply Difficulty: Medium

13. Award: **3.12 points**

Identify the sets that contain 2 as an element. (Check all that apply.)

- [ ] {2, 3, 4, 5, ...}
- [ ] {0, 1, 4, 9, ...}
- [ ] { 2, {2} }
- [ ] { {2}, { {2} } }
- [ ] { {2}, { 2, {2} } }
- [ ] { { {2} } }

References

Check All That Apply Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Find the cardinality of the set A.

References

Section Break Difficulty: Medium

14. Award: **3.12 points**

$A = \emptyset$

- ∞, because everything is in the empty set
- 1, as Ø is a single set
- -1, because nothing is in the empty set
- 0, as Ø is the empty set, which contains no elements

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Determine whether each of these sets is the power set of a set, where *a* and *b* are distinct elements.

References

Section Break Difficulty: Medium

15.

Award: **3.12 points**

{Ø, {*a*}, {Ø, *a*} }

- This is the power set of {Ø}.
- This is the power set of {a}.
- This is the power set of { a, Ø}
- This cannot be the power set of any set.

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $C = \{0, 1\}$.

References

Section Break Difficulty: Medium

16. Award: **3.12 points**

Identify $A \times B \times C$.

- {(*a*, *x*, *y*), (*a*, *x*, 1), (*a*, *y*, *x*), (*a*, *y*, 1), (*b*, *x*, *y*), (*b*, *x*, 1), (*b*, *y*, *x*), (*b*, *y*, 1), (*c*, *x*, *y*), (*c*, *x*, 1), (*c*, *y*, *x*), (*c*, *y*, 1)}
- {(0, *x*, *a*), (0, *x*, *b*), (0, *x*, *c*), (0, *y*, *a*), (0, *y*, *b*), (0, *y*, *c*), (1, *x*, *a*), (1, *x*, *b*), (1, *x*, *c*), (1, *y*, *a*), (1, *y*, *b*), (1, *y*, *c*)}
- {(0, *a*, *x*), (0, *a*, *y*), (0, *b*, *x*), (0, *b*, *y*), (0, *c*, *x*), (0, *c*, *y*), (1, *a*, *x*), (1, *a*, *y*), (1, *b*, *x*), (1, *b*, *y*), (1, *c*, *x*), (1, *c*, *y*)}
- {(*a*, *x*, 0), (*a*, *x*, 1), (*a*, *y*, 0), (*a*, *y*, 1), (*b*, *x*, 0), (*b*, *x*, 1), (*b*, *y*, 0), (*b*, *y*, 1), (*c*, *x*, 0), (*c*, *x*, 1), (*c*, *y*, 0), (*c*, *y*, 1)}

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider a set *A*.

References

Section Break Difficulty: Medium

17. Award: **3.12 points**

If $A = \{a\}$, then $A^3 = \{(a, a, a)\}$.

- True
- False

References

True / False Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Let *A* be the set of students who live within one mile of school and let *B* be the set of students who walk to classes. Describe the students in each of these sets.

References

Section Break Difficulty: Easy

18.

Award: **3.12 points**

$A \cup B$

- The set of students who walk to class
- The set of students who live within one mile of school and walk to class
- The set of students who live within one mile of school
- The set of students who live within one mile of school and the set of students who walk to class
- The set of students who either live within one mile of school or walk to class

References

Multiple Choice Difficulty: Easy

19.

Award: **3.12 points**

$B - A$

- The set of students who live within one mile of school but do not walk to class
- The set of students who live more than a mile from school but nevertheless walk to class
- The set of students who live within one mile of school except for the set of students who walk to class
- The set of students who walk to classes except for the set of students who live more than one mile of school.
- The set of students who live within one mile of school and the set of students who walk to class

References

Multiple Choice Difficulty: Easy

20.

Award: **3.12 points**

$A \cap B$

- The set of students who live within one mile of school or walk to class (or do both)
- The set of students who live within one mile of school
- The set of students who live within one mile of school and walk to class
- The set of students who walk to class
- The set of students who live within one mile of school and the set of students who walk to class

References

Multiple Choice Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find the value of given function.

References

Section Break Difficulty: Easy

21.

Award: **3.12 points**

$(A \cup B) \cap C$

- {2, 4, 6}
- {4, 5, 6, 8, 10}
- {4, 6}
- {0, 2, 4, 5, 6, 7, 8, 9, 10}

References

Multiple Choice Difficulty: Easy

22.

Award: **3.12 points**

Find the symmetric difference of {1, 3, 5} and {1, 2, 3}.

- {1, 3, 5, 9}

- {2}
- {2, 5}
- {1, 3}
- {5}

References

Multiple Choice Difficulty: Easy

23. Award: **3.12 points**

Identify the correct statements for the given functions from the set {*a*, *b*, *c*, *d*} to itself. (Check all that apply.)

- [ ] $f(a) = b, f(b) = a, f(c) = c, f(d) = d$ is a one-to-one function, as each element is an image of exactly one element.
- [ ] $f(a) = b, f(b) = a, f(c) = c, f(d) = d$ is not a one-to-one function, as *a* is mapped to *b* and *b* is mapped to *a*.
- [ ] $f(a) = b, f(b) = b, f(c) = d, f(d) = c$ is a one-to-one function, as each element is mapped to only one element.
- [ ] $f(a) = b, f(b) = b, f(c) = d, f(d) = c$ is not a one-to-one function, as *b* is an image of more than one element.
- [ ] $f(a) = d, f(b) = b, f(c) = c, f(d) = d$ is a one-to-one function, as each element is mapped to only one element.
- [ ] $f(a) = d, f(b) = b, f(c) = c, f(d) = d$ is not a one-to-one function, as *d* is an image of more than one element.

References

Check All That Apply Difficulty: Medium

24. Award: **3.12 points**

Which of the following functions from **Z** to **Z** are one-to-one? (Check all that apply.)

- [ ] $f(n) = n - 1$
- [ ] $f(n) = n^2 + 1$
- [ ] $f(n) = n^3$
- [ ] $f(n) = \lceil \frac{n}{2} \rceil$

References

Check All That Apply Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Consider the function *f*: **R** → **R** with $f(x) = ax + b$, where *a* and *b* are constants with $a \neq 0$.

References

Section Break Difficulty: Medium

25.

Award: **3.12 points**

Find the inverse of the given function.

- ○ $f^{-1}(x) = \frac{x}{a}$
- ○ $f^{-1}(x) = \frac{x+b}{a}$
- ○ $f^{-1}(x) = \frac{x-a}{b}$
- ○ $f^{-1}(x) = \frac{x-b}{a}$

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Let *f* be a function from the set *A* to the set *B*. Let *S* be a subset of *B*. We define the inverse image of *S* to be the subset of *A* whose elements are precisely all pre-images of all elements of *S*. We denote the inverse image of *S* by $f^{-1}(S)$, so $f^{-1}(S) = \{a \in A | f(a) \in S\}$. Let *f* be the function from **R** to **R** defined by $f(x) = x^2$. Find the value of given functions.

References

Section Break Difficulty: Medium

26. Award: **3.12 points**

$f^{-1}(\{1\})$
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- $f^{-1}(\{1\}) = \{-1,\ 1\}$
- $f^{-1}(\{1\}) = \{x\}$
- $f^{-1}(\{1\}) = \{-1\}$
- $f^{-1}(\{1\}) = \{1\}$

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Find these terms of the sequence $\{a_n\}$, where $a_n = 2(-3)^n + 5^n$.

References

Section Break Difficulty: Easy

27. Award: 3.12 points

a_0

- 0
- 3
- 1
- 5

References

Multiple Choice Difficulty: Easy

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
A person deposits $1,000 in an account that yields 9% interest compounded annually.

References

Section Break Difficulty: Medium

28. Award: 3.12 points

Set up a recurrence relation for the amount in the account at the end of n years.

- $a_n = 1.09a_{n-1}$

- $a_n = 0.09a_{n-1}$
- $a_n = a_{n-1} + 0.09$
- $a_n = a_{n-1} + 1.09$

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
For each of these lists of integers, find a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.

References

Section Break Difficulty: Medium

29. Award: **3.12 points**

2, 4, 16, . . .

- The next three terms are 256, 65536, 4294967296.
- The next three terms are 64, 2048, 262144.
- The next three terms are 256, 4096, 65536.
- The next three terms are 64, 256, 1024.

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
What are the values of these sums?

References

Section Break Difficulty: Medium

30. Award: **3.12 points**

$\sum_{k=1}^{5} (k+1) =$ _____

- 25
- 30
- 20
- 15

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
Find the value of each of the sums.

References

Section Break Difficulty: Medium

31. Award: **3.12 points**

$\sum_{j=0}^{8}\left(2^{j+1} - 2^{j}\right) =$ _____

- 411
- 500
- 576
- 511

References

Multiple Choice Difficulty: Medium

NOTE: *This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.*
The product of a_m, a_{m+1},...,a_n is represented by $\prod_{j=m}^{n} a_j$, read as the product from $j = m$ to $j = n$ of a_j. Identify the values of the given product.

References

Section Break Difficulty: Medium

32. Award: **3.28 points**

$\prod_{i=5}^{8} i$

References

Numeric Response Difficulty: Medium